

Reviewer Pack — Coxeter Group Action Count and Circuit Lower Bounds for Mono...

Entry e0990a55e08d · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Coxeter Group Action Count and Circuit Lower Bounds for Monotone k-CLIQUE

Entry ID: e0990a55e08d **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-21 12:25:06 UTC

1. Conjecture

Field A (mathematical branch): Combinatorial Group Theory **Field B** (complexity object): Boolean Circuit Complexity

Statement:

[‘The number of distinct orbits under the action of a Coxeter group on the set of vertices of a monotone k-CLIQUE DNF is upper-bounded by a polynomial in k.’, ‘For all monotone k-CLIQUE formulas, there exists a circuit with size at most $O(k^2)$ that computes the OR of all satisfying assignments of the formula.’, ‘If there exists a monotone k-CLIQUE formula that requires a circuit of size more than $n^{2.5}$ for its computation, then the action of the Coxeter group on the set of vertices contains an orbit with at least 10^6 elements.’]

Rationale (proposer’s reasoning):

[‘Coxeter groups provide a natural framework to study symmetries and classify objects in combinatorial structures.’, ‘If the number of orbits under the action of a Coxeter group is polynomially bounded, it implies a structural simplicity that might lead to efficient algorithms for computing monotone k-CLIQUE formulas.’, ‘This conjecture connects the algebraic properties of Coxeter groups with the complexity of Boolean circuits, potentially revealing new insights into the hardness of monotone k-CLIQUE.’]

Taxonomy category: KARCHMER_WIGDERSON (status at proposal time: alive)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 17306bbf5ebaec88

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

Counting distinct orbits under Coxeter group action for monotone k-CLIQUE DNFs and comparing to polynomial upper bound in terms of k.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - "Coxeter group action" AND "monotone k-CLIQUE DNF" AND "polynomial upper bound" - "Boolean circuit complexity" AND " $0(k^2)$ size" AND "satisfying assignments" - "circuit lower bounds" AND "monotone k-CLIQUE" AND "Coxeter group orbit"

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=1, elapsed=0.1s

5.1 Generated Python source

```
import random
import math

def generate_monotone_k_clique(n, k):
    if n < k:
        return None
    vertices = list(range(n))
    clique_edges = set()
    for i in range(k):
        for j in range(i + 1, k):
            clique_edges.add((vertices[i], vertices[j]))
    remaining_edges = [(i, j) for i in range(k, n) for j in range(i + 1, n)]
    random.shuffle(remaining_edges)
    for u, v in remaining_edges[:n - k]:
        if (u, v) not in clique_edges and (v, u) not in clique_edges:
            clique_edges.add((u, v))
    return clique_edges

def generate_coxeter_group_action(n):
    # Simplified Coxeter group action for demonstration
    # This is a placeholder and should be replaced with actual group theory code
    action = {}
    for i in range(n):
        action[i] = (i + 1) % n
    return action

def count_distinct_orbits(clique_edges, action):
    orbits = set()
    visited = set()
    for v in clique_edges:
```

```

    if v not in visited:
        orbit = {v}
        current = v
        while True:
            next_v = action[current]
            if next_v == v:
                break
            orbit.add(next_v)
            current = next_v
        orbits.add(frozenset(orbit))
        visited.update(orbit)
    return len(orbits)

def run_trial(seed: int) -> dict:
    random.seed(seed)
    n_values = [5, 10, 15, 20, 30, 40]
    results = []

    for n in n_values:
        k = min(n // 2, 10) # Ensure k is at least 1 and at most n//2
        clique_edges = generate_monotone_k_clique(n, k)
        if clique_edges is None:
            continue

        action = generate_coxeter_group_action(n)
        orbits_count = count_distinct_orbits(clique_edges, action)

        results.append({
            "n": n,
            "k": k,
            "orbits_count": orbits_count
        })

    if not results:
        return {
            "metric_name": "Orbits Count",
            "metric_value": 0,
            "instances_tested": 0,
            "conjecture_holds": False,
            "counterexample": "No valid monotone k-CLIQUE formula generated"
        }

    mean_orbits = sum(result["orbits_count"] for result in results) / len(results)
    std_orbits = math.sqrt(sum((result["orbits_count"] - mean_orbits) ** 2 for result in results) / len(results))

    return {
        "metric_name": "Orbits Count",
        "metric_value": mean_orbits,
        "instances_tested": len(results),
        "conjecture_holds": mean_orbits <= k**3, # Polynomial upper bound
        "counterexample": ""
    }

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] or [2, 3, 5, 7, 11, 13, 17, 19, 23, 29] * 3

```

```

results = []
for seed in seeds:
    result = run_trial(seed)
    print(f"TRIAL: {result}")
    results.append(result)

mean_orbits = sum(r["metric_value"] for r in results) / len(results)
std_orbits = math.sqrt(sum((r["metric_value"] - mean_orbits) ** 2 for r in results) / len(results))
support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

if all(r["conjecture_holds"] for r in results):
    print(f"RESULT: SUPPORTED mean={mean_orbits} std={std_orbits} support_fraction={support_fraction}")
elif support_fraction >= 0.8:
    print(f"RESULT: SUPPORTED mean={mean_orbits} std={std_orbits} support_fraction={support_fraction}")
else:
    for r in results:
        if not r["conjecture_holds"]:
            counterexample = f"n={r['n']}, k={r['k']}, orbits_count={r['orbits_count']}"
            print(f"RESULT: FALSIFIED counterexample={counterexample} first_failing_seed={seed}")
            break

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_2075bdeb.py", line 104, in <module>
    result = run_trial(seed)
              ~~~~~^~~~~~

  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_2075bdeb.py", line 70, in run_trial
    orbits_count = count_distinct_orbits(clique_edges, action)
                    ~~~~~^~~~~~

  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_2075bdeb.py", line 49, in count_distinct_orbits
    next_v = action[current]
              ~~~~~^~~~~~
KeyError: (0, 1)

```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test crashed before producing data, which means that the pre-registered support condition could not be unambiguously met. | next: Re-run the test to ensure it completes successfully and verify if the pre-registered support condition is met.

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	16258
2	preregistration	ollama_remote	glm4:latest	0	0	10096
3	novelty	ollama_remote	glm4:latest	0	0	9574
4	novelty	ollama_remote	glm4:latest	0	0	10124
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	11223
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	8553
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	10250
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	11827
9	judge	ollama_remote	glm4:latest	0	0	12083

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 99988 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in research/audit/e0990a55e08d.jsonl)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in research/audit/e0990a55e08d.jsonl for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: research/pvsnp_sandbox/test_*.py (per-cycle)

Replay tarball: research/replay/e0990a55e08d.tar.gz (if generated)